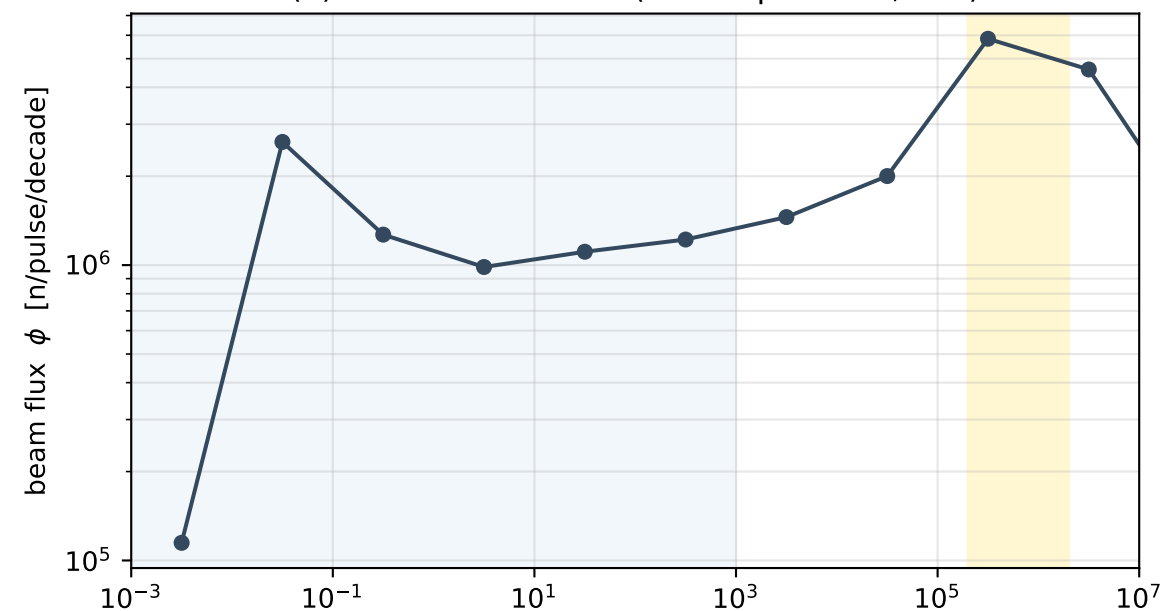
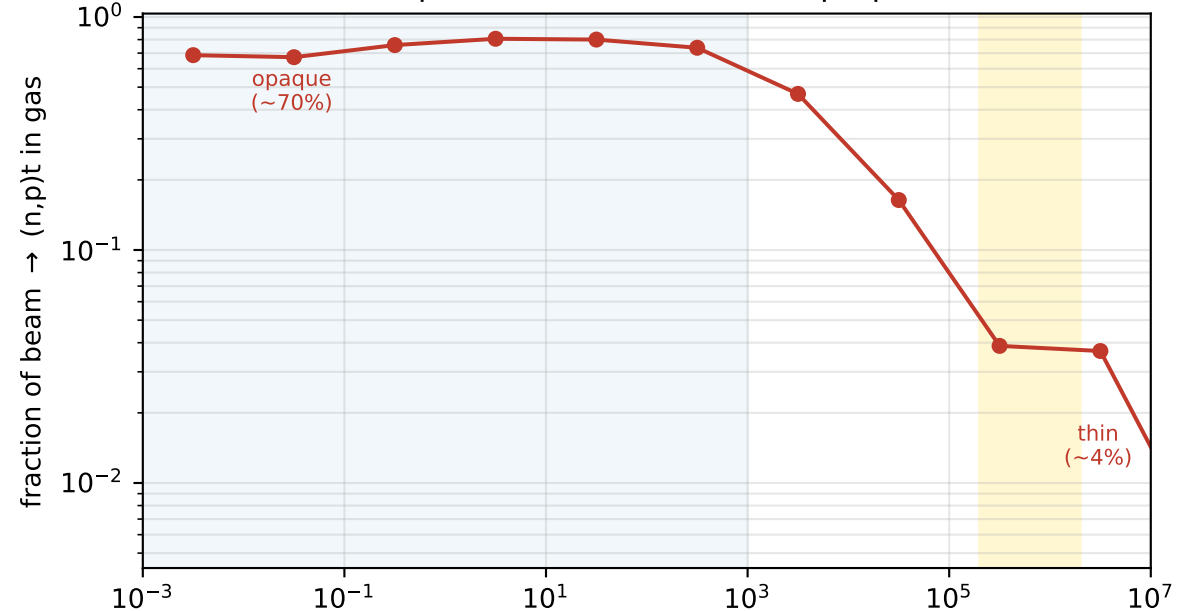


From cross section to statistics: $N_X/\text{day} = \phi \times f_{\text{breakup}} \times (\sigma_X/\sigma_{np}) \times \text{pulses/day}$ (gold = 0.2-2 MeV window, blue = sub-keV)

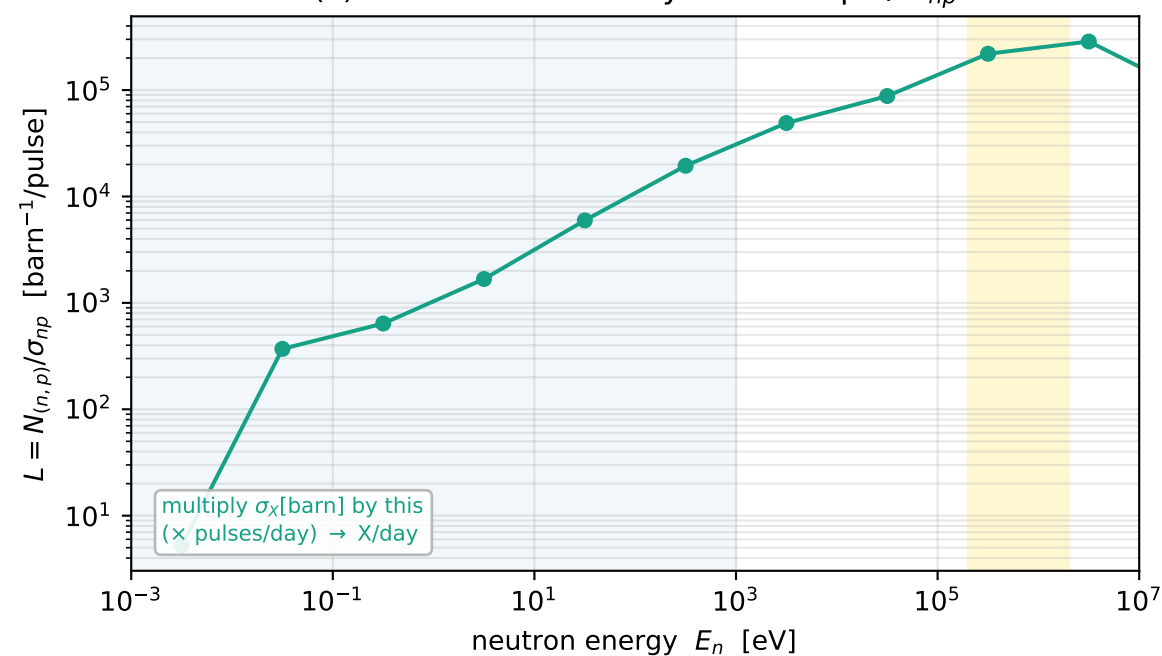
(a) total neutron flux (EAR2 spectrum, sim)



(b) breakup (interaction) fraction: opaque $\rightarrow$ thin



(c) effective luminosity = breakups / σ_{np}



(d) background burden: (n,p)t breakups per X17 produced

